Summary Tables for Chemistry Unit 1 (Edexcel IAL)

1. Atomic Radius

Direction	Trend	Explanation	Exceptions
Across a Period	Decreases	As you move from left to right across a period, the number of protons in the nucleus increases, leading to a higher effective nuclear charge. This increased attraction pulls the electron cloud closer to the nucleus, resulting in a smaller atomic radius.	The noble gases at the end of each period have a full valence shell, leading to a slightly larger atomic radius than the preceding halogens due to increased electron-electron repulsion in the filled shell.
Down a Group	Increases	Moving down a group, each successive element has an additional electron shell, increasing the distance between the nucleus and the outermost electrons, thereby increasing the atomic radius.	Transition metals (not needed for Unit1)

2. Ionization Energy

Direction	Trend	Explanation	Exceptions
Across a Period	Increases	From left to right across a period, the effective nuclear charge increases, making it more difficult to remove an electron, thus increasing ionization energy.	There are slight decreases between groups 2 and 13 (e.g., Be to B) and groups 15 and 16 (e.g., N to O). In the former, the electron removed from group 13 elements is from a p-orbital, which is higher in energy than the s-orbital of group 2. In the latter, the paired electrons in the p-orbital of group 16 elements experience electron-electron repulsion, making them easier to remove.
Down a Group	Decreases	As you move down a group, the outer electrons are farther from the nucleus due to additional electron shells, and increased shielding reduces the effective nuclear charge felt by these electrons, making them easier to remove.	Noble gases have exceptionally high ionization energies due to their stable electron configurations.

3. Electronegativity

Direction	Trend	Explanation	Exceptions
Across a Period	Increases	The increasing effective nuclear charge across a period attracts bonding electrons more strongly, leading to higher electronegativity.	The noble gases are often excluded from electronegativity trends because they typically do not form bonds.
Down a Group	Decreases	Additional electron shells increase the distance between the nucleus and bonding electrons, reducing the nucleus's pull and thus decreasing electronegativity.	The decrease in electronegativity down a group is less pronounced in the transition metals due to their complex electron configurations. (not needed for unit 1)

4. Bond shape and angle

Bond Angle	Shape	Number of Lone Pairs	Number of Bond Pairs	Example
180°	Linear	0	2	CO_2
120°	Trigonal Planar	0	3	BF_3
109.5°	Tetrahedral	0	4	CH_4
107°	Trigonal Pyramidal	1	3	NH_3
104.5°	Bent (V-shape)	2	2	H_2O
90° & 120°	Trigonal Bipyramidal	0	5	PCl_5
90°	Octahedral	0	6	SF_6
89°	Square Pyramidal	1	5	BrF_5
90°	Square Planar	2	4	XeF_4

Reason

Molecules with lone pairs: Lone pairs repel more than bond pairs. The electron pairs arrange themselves in positions of maximum separation and minimum repulsion.

Molecules with no lone pairs: The bonding regions of electrons or bond pairs repel equally. The bonding pairs arrange themselves in positions of maximum separation and minimum repulsion.

5. Functional groups

Functional Group	Name
-C-C-	Alkane
C=C	Alkene
-C≡C-	Alkyne
-OH	Alcohol
-COOH	Carboxylic Acid
-CHO	Aldehyde
-C=O	Ketone
-COO-	Ester
$-NH_2$	Amine
$-CONH_2$	Amide
-CN	Nitrile
$-NO_2$	Nitro Compound
-X (F, Cl, Br, I)	Halogenoalkane

6. Reactions of alkanes

Reaction Type	Description	Conditions	Example Equation	Key Notes
Combustion	Alkanes react with oxygen to produce carbon dioxide and water (complete combustion). Incomplete combustion produces carbon monoxide or soot.	Excess O_2, ignition (e.g., spark or flame)	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	Exothermic; incomplete combustion occurs with limited O_2.
Substitution (Free Radical)	Halogens replace hydrogen atoms in the presence of UV light free radical mechanism.	UV light, halogen (e.g., Cl_2 or Br_2)	$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$	Proceeds via initiation, propagation, and termination steps. Multiple products possible.
Cracking	Large alkanes break into smaller alkanes and alkenes.	High temperature (500–700°C), catalyst (e.g., zeolite)	$C_{10}H_{22} \rightarrow C_5H_{12} + C_5H_{10}$	Used in industry to produce smaller, useful hydrocarbons.
Reforming	Straight-chain alkanes are converted into branched or cyclic hydrocarbons.	High temperature, catalyst (e.g., platinum)	$C_6H_{14} \rightarrow C_6H_{12} + H_2$	Improves fuel quality (e.g., octane rating).

7. Reactions of alkenes

Reaction Type	Description	Conditions	Example Equation	Key Notes
Addition (Hydrogenation)	Hydrogen adds across the double bond via **catalytic addition**, forming an alkane.	H_2 and Ni or Pt catalyst, heat (150°C)	$CH_2=CH_2 + H_2 \rightarrow CH_3CH_3$	Mechanism: Catalytic addition; used industrially (e.g., margarine).
Addition (Halogenation)	Halogens add across the double bond via **electrophilic addition**, forming a dihaloalkane.	Room temp, no catalyst needed	$CH_2=CH_2 + Br_2 \rightarrow CH_2BrCH_2Br$	Electrophilic addition; Br_2 decolorizes (test for unsaturation).
Addition (Hydrogen Halides)	HX adds across the double bond via **electrophilic addition**.	Room temp, no catalyst	$CH_2=CH_2 + HBr \rightarrow CH_3CH_2Br$	Electrophilic addition; follows Markovnikov's rule for unsymmetrical alkenes.
Addition (Water - Hydration)	Water adds across the double bond via **electrophilic addition** to form an alcohol.	H_3PO_4 catalyst, steam, 300°C, 60 atm	$CH_2=CH_2 + H_2O \rightarrow CH_3CH_2OH$	Acid-catalyzed electrophilic addition; Markovnikov applies.
Oxidation (Cold $KMnO_4$)	Cold, dilute $KMnO_4$ oxidizes the double bond to a diol	Cold, dilute $KMnO_4$, OH^- (aqueous NaOH)	$CH_2=CH_2 + [O] + H_2O \rightarrow CH_2OHCH_2OH$	Oxidation reaction; purple $KMnO_4$ decolorizes (unsaturation test).
Oxidation (Hot $KMnO_4$)	Hot, concentrated $KMnO_4$ cleaves the double bond via **oxidative cleavage**, forming ketones, aldehydes, or CO_2.	Hot, conc. $KMnO_4$, H^+ (sulfuric acid)	$CH_3CH=CH_2 \rightarrow CH_3COOH + CO_2$	Oxidative cleavage; products depend on alkene structure.
Polymerisation	Alkenes join via **free radical addition polymerisation** to form long chains.	High pressure, catalyst (e.g., Ziegler-Natta)	$nCH_2=CH_2 \rightarrow [–CH_2CH_2–]_n$	Mechanism: Free radical addition; forms polymers like polyethene.

Notes

1. Electrophilic Addition

Alkenes react with electrophiles (e.g., Br_2, HBr) due to the electron-rich C=C double bond. The π-bond breaks, and new σ-bonds form across the carbons.

Step 1: The electrophile (e.g., H^+ from HBr or Br^+ from Br_2) is attracted to the π-electrons of the double bond. This forms a **carbocation intermediate**.

  - Example: $CH_2{=}CH_2 + HBr \rightarrow [CH_3CH_2^+] + Br^-$

- **Step 2**: The nucleophile (e.g., Br^-) attacks the carbocation, forming the product. The nucleophile adds to the carbocation
  - $CH_3CH_2^+ + Br^- \rightarrow CH_3CH_2Br$

2. Free Radical Substitution

Alkanes react with halogens (e.g., Cl_2) under UV light, replacing H with a halogen via a free radical mechanism.

Mechanism

- **Initiation**: UV light breaks the Cl–Cl bond homolytically, forming free radicals.
  - $Cl_2 \rightarrow 2Cl\bullet$ (UV light)
- **Propagation**: Radicals react in a chain reaction.
  - $Cl\bullet + CH_4 \rightarrow CH_3\bullet + HCl$
  - $CH_3\bullet + Cl_2 \rightarrow CH_3Cl + Cl\bullet$
- **Termination**: Radicals combine, stopping the chain.
  - $Cl\bullet + Cl\bullet \rightarrow Cl_2$
  - $CH_3\bullet + Cl\bullet \rightarrow CH_3Cl$
  - $CH_3\bullet + CH_3\bullet \rightarrow C_2H_6$

3. Markovnikov's Rule

When an unsymmetrical alkene (e.g., propene, $CH_3CH{=}CH_2$) reacts with HX (e.g., HBr), the hydrogen adds to the carbon with more hydrogen atoms already attached, and the halogen adds to the carbon with fewer hydrogens. This is due to carbocation stability.
Example: **propene ($CH_3CH{=}CH_2$) + HBr**.

Step 1: Electrophile Attack

⇨ The π-electrons of the C=C bond attack the electrophile (H^+ from HBr), breaking the double bond.
⇨ Two possible carbocations can form:

Option 1: H^+ adds to C1 ($CH_2=$), forming [$CH_3CH^+CH_3$] (secondary carbocation).

Option 2: H^+ adds to C2 (=CH–), forming [$CH_3CH_2CH_2^+$] (primary carbocation).

Step 2: Nucleophile Attack

- The nucleophile (Br^-) adds to the carbocation.

⇨ From [$CH_3CH^+CH_3$] → $CH_3CHBrCH_3$ (2-bromopropane).
⇨ From [$CH_3CH_2CH_2^+$] → $CH_3CH_2CH_2Br$ (1-bromopropane).

- **Result**: 2-bromopropane is the **major product** (Markovnikov product); 1-bromopropane is the **minor product** (anti-Markovnikov product). The major product is mainly formed as it is more stable than the minor product which may form in very small or trace amounts.

Definitions

Topic 1: Formulae, Equations, and Amount of Substance

Term	Definition
Atom	The smallest part of an element that can exist. All substances are made up of atoms.
Atom economy	Measure of the proportion of reacting atoms that become part of the desired product in the balanced chemical equation.
Avogadro constant (L)	The number of atoms, molecules, or ions in one mole of a given substance. It is the number of atoms in exactly 12 g of carbon-12 (6.02×10^{23} mol^{-1}).
Compound	A substance that combines two or more different elements through the formation of chemical bonds.
Element	A substance that contains one type of atom only. Atoms of the same element all have the same number of protons.
Empirical formula	Smallest whole number ratio of atoms of each element in a compound.
Ion	Formed when an atom loses or gains electrons, giving it an overall charge.
Ionic equation	A chemical equation that involves dissociated ions.
Molar mass	Mass of one mole of a substance expressed in g/mol.
Mole	The unit for the amount of substance. One mole is 6.02×10^{23} particles.
Molecular formula	The actual number of atoms of each element in a molecule.
Molecule	Formed from two or more atoms held together by covalent bonds.
Percentage yield	The percentage ratio of the actual yield of a product from a reaction compared with the theoretical yield.
Relative atomic mass	Average mass of an atom of an element relative to 1/12th of the mass of an atom of carbon-12.
Relative formula mass	Average mass of a compound relative to 1/12th of the mass of an atom of carbon-12 (for giant structures).
Relative molecular mass	Average mass of a molecule relative to 1/12th of the mass of an atom of carbon-12.
State symbol	Shows the physical state of a substance in a reaction: gas (g), liquid (l), solid (s), aqueous (aq).

Topic 2: Atomic Structure and the Periodic Table

Term	Definition
Atomic/Proton number	The number of protons in the nucleus of an atom of a particular element.
Boiling temperature	The temperature at which a substance changes from a liquid to a gas.
Diatomic molecule	A molecule containing two atoms chemically bonded together (e.g., Cl_2).
Electron	Negatively charged subatomic particle orbiting the nucleus.
Electron subshell	Divisions of electron shells at different energy levels (s, p, d, f).
Electronic configuration	The distribution of electrons in an atom's orbitals.
Endothermic reaction	A reaction that absorbs energy from the surroundings, causing a temperature drop.
First ionization energy	The energy required to remove one mole of electrons from one mole of gaseous atoms to form gaseous 1+ ions.
Isotope	Atoms of the same element with the same number of protons but different numbers of neutrons.
Mass number	Sum of protons and neutrons in an atom's nucleus.
Mass spectrometry	A technique measuring the mass-to-charge ratio of gaseous ions.
Melting temperature	The temperature at which a solid turns into a liquid.
Neutron	Neutral subatomic particle in the nucleus with a relative mass of 1.
Nuclear charge	Total charge of all protons in the nucleus.
Orbital	A region around a nucleus that can hold up to two electrons.
p orbital	A dumbbell-shaped orbital that can hold up to six electrons in total.
Periodicity	Trends in properties of elements across periods due to atomic structure.
Proton	Positively charged subatomic particle in the nucleus.
Quantum shell	Electron energy levels assigned a principal quantum number (n).
Relative abundance	The percentage of isotopes of an element in a naturally occurring sample.
Relative peak height	Found in mass spectra, indicating the relative abundance of a substance.
s orbital	Spherical orbital that can hold up to two electrons.

Second ionization energy	Energy required to remove one mole of electrons from 1+ gaseous ions to form 2+ gaseous ions.
Shielding	Decrease in nuclear attraction experienced by an outer electron due to inner electron repulsion.
Subatomic particles	Particles smaller than an atom (protons, neutrons, electrons).
Third ionization energy	Energy required to remove one mole of electrons from 2+ gaseous ions to form 3+ gaseous ions.

Topic 3: Bonding and Structure

Term	Definition
Anion	A negatively charged ion formed by gaining electrons.
Atomic nucleus	The positively charged center of an atom, containing protons and neutrons.
Bond angle	The angle between two bonds in a molecule.
Bond length	The distance between two covalently bonded atoms.
Cation	A positively charged ion formed by losing electrons.
Covalent bond	Strong electrostatic attraction between two nuclei and shared electron pairs.
Covalent substance	A substance made of atoms held together by covalent bonds.
Dative covalent bond	A covalent bond where one atom donates both electrons.
Delocalized electrons	Electrons not confined to a single atom or bond.
Dot-and-cross diagram	A representation of bonding in molecules using dots and crosses.
Double bond	A covalent bond involving four shared electrons.
Electrical conductivity	A measure of how well a substance carries electric current.
Electron density map	A visual representation of electron distribution in a molecule.
Electron pair repulsion	The repulsion between electron pairs determines molecular shape.
Electronegativity	An atom's ability to attract bonding electrons.
Electrostatic attraction	The force between oppositely charged particles.
Giant atomic structure	A large, covalently bonded lattice (e.g., diamond).
Giant ionic lattice	A regular, repeating structure of oppositely charged ions.
Ionic bond	Strong electrostatic attraction between oppositely charged ions.

Ionic charge	The charge on an ion due to electron gain or loss.
Ionic compound	A compound made of ions, held together by ionic bonds.
Ionic radius	The distance from the nucleus to the outermost electron of an ion.
Isoelectronic ions	Ions or atoms with the same electron configuration.
Metallic bonding	Attraction between positive metal ions and delocalized electrons.
Polar bond	A covalent bond where electrons are unequally shared.
Polar molecule	A molecule with uneven electron distribution, creating dipoles.
Polarisability	The ability of a molecule to induce a dipole in another.
Single bond	A covalent bond involving two shared electrons.
Triple bond	A covalent bond involving six shared electrons.

Topic 4: Introductory Organic Chemistry and Alkanes

Term	Definition
Addition	Joining two or more molecules together to form a larger molecule. Hydration is the addition of an H_2O molecule. Halogenation involves the addition of a halogen. Hydrogenation is the addition of H_2. Electrophilic addition describes all these processes.
Alkane	A homologous series of saturated hydrocarbons with the general formula C_nH_{2n+2}.
Alkane fuel	Alkanes are good fuels as they burn cleanly, undergoing combustion in excess oxygen to produce only carbon dioxide and water.
Bond breaking	When a bond breaks, energy is absorbed, making it an endothermic process.
Carbon neutrality (fuels)	A fuel is carbon neutral if its production and use do not lead to a net increase in atmospheric carbon dioxide.
Catalyst	A substance that speeds up a reaction by providing an alternative pathway with lower activation energy. Catalysts are not consumed in the reaction.
Chain isomers	Structural isomers that have different carbon chain arrangements due to branching.
Combustion of alkanes	The burning of alkanes to release energy. In complete combustion, alkanes produce only CO_2 and H_2O. In incomplete combustion, carbon monoxide and carbon particulates can form.
Cracking	Breaking long-chain alkanes into smaller, more useful hydrocarbons, typically to produce shorter alkanes and alkenes.
Crude oil	A finite resource found in rocks, made up mainly of hydrocarbons derived from ancient biomass.
Cycloalkane	Saturated hydrocarbons with a cyclic ring structure. They contain only single bonds and have the general formula C_nH_{2n}.
Electrophile	An electron-pair acceptor in an organic reaction, attracted to regions of high electron density.
Fractional distillation	A method of separating a mixture of liquids by differences in boiling points using a fractionating column.
Free radical substitution	A reaction between alkanes and halogens under UV light, forming halogenoalkanes. Involves initiation, propagation, and termination steps.
Functional group	A specific group of atoms responsible for the characteristic reactions of a compound.

Functional group isomers	Structural isomers that belong to different homologous series due to differing functional groups.
Hazard	A property of a chemical that could cause harm.
Heterolytic bond breaking	Bond breaking where both electrons move to one atom, forming a cation and an anion.
Homologous series	A series of organic compounds with the same functional group and general formula. Members differ by a CH_2 unit.
Homolytic bond breaking	Bond breaking where electrons are split evenly between two atoms, forming radicals.
Hydrocarbon	A compound consisting only of hydrogen and carbon atoms.
Initiation step	The first step in radical substitution, where radicals are generated via homolytic bond cleavage.
Mechanism	A step-by-step representation of electron movement in a reaction.
Organic molecule	A molecule containing carbon and hydrogen, often with additional elements like oxygen or nitrogen.
Oxidation	Loss of electrons or an increase in oxidation state.
Pollutant	A chemical substance that harms the environment.
Polymerisation	A reaction in which monomers join together to form a polymer.
Position isomer	Structural isomers differing in the position of a functional group.
Propagation step	The stage in a free radical substitution reaction where radicals react to form new radicals.
Radical	A species with an unpaired electron, represented with a dot (•).
Reduction	Gain of electrons or a decrease in oxidation state.
Risk	The likelihood of harm from a hazard.
Risk assessment	Evaluating potential risks when handling chemicals or conducting experiments.
Substitution reaction	A reaction where one atom or group is replaced by another.
Structural isomers	Compounds with the same molecular formula but different structural arrangements.
Synthesis	The process of combining different elements or compounds to form new molecules.
Termination	The final step in a radical substitution mechanism, where two radicals combine to form a stable molecule.

Thermal decomposition	A reaction in which a substance breaks down upon heating.
Toxicity	A measure of how harmful a chemical is to organisms.

Topic 5: Alkenes

Term	Definition
Biodegradable polymer	A polymer that can be broken down by bacteria or other organisms.
Cycloalkenes	Unsaturated hydrocarbons containing a cyclic ring with both single and double bonds.
Geometric isomerism	A type of stereoisomerism due to restricted rotation around a carbon-carbon double bond. In Z-isomers, the highest priority groups are on the same side; in E-isomers, they are on opposite sides.
Polymer	A long-chain molecule formed from repeating monomer units.
Polymer disposal	Most polymers are non-biodegradable. They can be disposed of in landfills, incinerated (producing greenhouse gases), or recycled.
Primary carbocation	A carbocation where the positively charged carbon is bonded to one alkyl group. Least stable.
Secondary carbocation	A carbocation where the positively charged carbon is bonded to two alkyl groups. More stable than a primary carbocation.
Tertiary carbocation	A carbocation where the positively charged carbon is bonded to three alkyl groups. Most stable.
Unsaturated hydrocarbon	A hydrocarbon that contains at least one carbon-carbon double bond, consisting of a sigma (σ) bond and a pi (π) bond.

Note: This is only a summary for revision; calculations, mass spectroscopy or polarization is not covered in details. For detailed understanding, check this playlist out: https://www.youtube.com/playlist?list=PLu2oZOKkN0kcBvsSJI-ZGEboGBg3TpfXc